

spread locator

Topic	Purpose
Bernoulli Distribution	Model binary events
Binomial Distribution	Count successes in multiple trials
Poisson Distribution	Model event frequency in time
Log-Normal Distribution	Model positive skewed financial data
Power Law Distribution	Model heavy-tailed large values
Q-Q Plot	Test normality of data
Box-Cox Transformation	Reduce skewness
Z-Score	Standardize values
Probability using Z	Compute probability from normal distribution
PDF	Density of probability
CDF	Cumulative probability

Assume a **sample transaction dataset (10 days)**:

Transaction Occurrence (1=Yes,0=No):

1,0,1,1,0,1,1,0,1,1

Weekly transaction counts (7 days):

4,5,3,6,4,5,3

Transactions per day:

3,4,2,5,3,4,6,2,3,5

Transaction Amounts (₹):

1200, 2500, 3200, 1500, 4100, 2300, 5200, 6100, 4800, 3500

Total observations = **10**

1. Bernoulli Distribution (Transaction Occurrence)

Bernoulli distribution models **binary outcomes (0 or 1)**.

Step 1: Probability of transaction occurring

Number of transactions = **7**

Total observations = **10**

$$p = \frac{7}{10}$$
$$p = 0.7$$

Step 2: Bernoulli Probability

Bernoulli PMF:

$$P(X = x) = p^x(1 - p)^{1-x}$$

For transaction occurrence $X = 1$:

$$P(X = 1) = p$$
$$P(X = 1) = 0.7$$

For no transaction $X = 0$:

$$P(X = 0) = 1 - p$$
$$P(X = 0) = 0.3$$

2. Binomial Distribution (Weekly Transaction Count)

Weekly transaction counts:

4,5,3,6,4,5,3

Total trials per week:

$$n = 7$$

Probability of transaction:

$$p = 0.7$$

Probability of exactly 4 transactions

Binomial Formula

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

$$P(X = 4) = \binom{7}{4} (0.7)^4 (0.3)^3$$

Step 1

$$\binom{7}{4} = \frac{7!}{4! 3!} = 35$$

Step 2

$$(0.7)^4 = 0.2401$$

Step 3

$$(0.3)^3 = 0.027$$

Step 4

$$P(X = 4) = 35 \times 0.2401 \times 0.027$$

$$P(X = 4) = 35 \times 0.00648$$

$$P(X = 4) = 0.2268$$

3. Poisson Distribution (Transactions per Day)

Daily transactions:

3,4,2,5,3,4,6,2,3,5

Step 1: Mean (λ)

$$\lambda = \frac{\sum x}{n}$$

Sum:

$$3 + 4 + 2 + 5 + 3 + 4 + 6 + 2 + 3 + 5 = 37$$

$$\lambda = \frac{37}{10}$$

$$\lambda = 3.7$$

Step 2: Probability of 4 transactions

Poisson formula

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$$

$$P(X = 4) = \frac{e^{-3.7} (3.7)^4}{4!}$$

Step 1

$$e^{-3.7} = 0.0247$$

Step 2

$$3.7^4 = 187.4$$

Step 3

$$4! = 24$$

Step 4

$$P(X = 4) = \frac{0.0247 \times 187.4}{24}$$

$$P(X = 4) = \frac{4.628}{24}$$

$$P(X = 4) = 0.193$$

4. Log-Normal Distribution (Transaction Amounts)

Amounts (₹)

1200,2500,3200,1500,4100,2300,5200,6100,4800,3500

Step 1: Take Natural Log

$$y = \ln(x)$$

Example

$$\ln(1200) = 7.09$$

$$\ln(2500) = 7.82$$

$$\ln(3200) = 8.07$$

(continue similarly)

Step 2: Mean of log values

Assume

$$\mu = 8.05$$

Standard deviation

$$\sigma = 0.45$$

Thus

$$X \sim \text{LogNormal}(8.05, 0.45)$$

5. Q-Q Plot Normality Test (Mathematical idea)

Ordered data:

1200,1500,2300,2500,3200,3500,4100,4800,5200,6100

Step 1

Calculate **sample quantiles**

Step 2

Calculate **normal quantiles**

Step 3

Plot

(*sample q*

quantiles, theoretical q

If points follow straight line → data approximately normal.

6. Box-Cox Transformation

Formula

$$y(\lambda) = \frac{x^\lambda - 1}{\lambda}$$

Assume

$$\lambda = 0.5$$

Example for ₹1200

$$\begin{aligned} y &= \frac{1200^{0.5} - 1}{0.5} \\ \sqrt{1200} &= 34.64 \\ y &= \frac{34.64 - 1}{0.5} \\ y &= 67.28 \end{aligned}$$

7. Z-Score Calculation

Formula

$$Z = \frac{x - \mu}{\sigma}$$

Step 1: Mean

$$\mu = \frac{\sum x}{n}$$

Sum

$$\begin{aligned} &1200 + 2500 + 3200 + 1500 + 4100 + 2300 + 5200 + 6100 + 4800 + 3500 \\ &= 34400 \\ \mu &= \frac{34400}{10} \\ \mu &= 3440 \end{aligned}$$

Step 2: Standard Deviation

Assume

$$\sigma = 1500$$

Step 3: Z-score for ₹5000

$$\begin{aligned} Z &= \frac{5000 - 3440}{1500} \\ Z &= \frac{1560}{1500} \\ Z &= 1.04 \end{aligned}$$

Step 4: Probability

From Z table

$$P(Z < 1.04) = 0.8508$$

Probability transaction exceeds ₹5000

$$P(X > 5000) = 1 - 0.8508$$
$$P(X > 5000) = 0.1492$$

8. PDF and CDF

PDF Formula

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Example

$$f(5000) = \frac{1}{1500\sqrt{2\pi}} e^{-\frac{(5000-3440)^2}{2(1500)^2}}$$

CDF

$$F(x) = P(X \leq x)$$

Example

$$F(5000) = 0.8508$$

Meaning **85.08% transactions are $\leq$ ₹5000**